

Is Generative AI Displacing Arabic? Perceived Change in Arabic Use Among Arabic–English Bilinguals in Lebanon

Hadi Ghazi
Lebanese University
Lebanon
hadigghazi@gmail.com

Abstract—Generative AI works better in English than in Arabic, so bilinguals who rely on it may route more of their work through English. In an anonymous Arabic online survey of 40 bilingual AI users (primary sample: 37 who grew up in the Arab region and live in Lebanon), respondents rated whether, because of AI, they now use Arabic less or more in eight domains. Most reported AI-attributed decreases in work or study (25 of 36, none reporting more) and in writing messages, and decreases outnumbered increases for formal Arabic (16 against 3). The decreases followed the order predicted by domain theory, from work and study to personal matters to family, an order that may also reflect where AI is used, while the predicted larger decrease for formal than for dialect domains was not found. How intensively people use AI, the language they use with it and the quality gap they perceive did not jointly predict the reported decrease. In cross-validated machine learning, AI-use information added no detectable predictive value, and no usable model emerged. Oversampling inside the training folds did not help; applied before the train–test split, it produced spurious models with an apparent area under the curve of up to 0.96. The AI attribution went together with blaming social media. Respondents attribute to AI a decrease in their Arabic, but the data cannot show that AI causes it, nor separate the AI attribution from a broader sense that technology is eroding Arabic.

Index Terms—generative AI, Arabic, language shift, diglossia, perceived language change, survey, machine learning

I. INTRODUCTION

Generative AI tools perform less well in Arabic than in English: tool-calling accuracy drops when the same tasks are prompted in Arabic [1], and four chatbots answered the same infectious-disease questions less well in Arabic than in English [2]. For bilinguals who use these tools daily, a task routed through AI may therefore also be routed through English. Arabic was already losing formal domains to English before generative AI: English dominates higher education in the UAE [3], where students at English-medium universities report weaker academic Arabic [4]; Qatar University students rate English more useful than Arabic for scientific and professional communication [5]; and in Lebanon, a small interview study found English used across education and professional life [6].

Research on generative AI and its users’ language skills has looked almost entirely at second-language learning, e.g., [7]. The closest work on Arabic records which language Tunisian users choose with ChatGPT [8], not whether that

choice changes their Arabic elsewhere. To our knowledge, no study has asked Arabic speakers whether they attribute changes in their own Arabic use to AI.

This study asks Arabic–English bilinguals in Lebanon in which domains, and in which direction, they report that AI has changed their use of Arabic (RQ1); whether AI-use intensity, the English share of AI use and the perceived Arabic–English quality gap predict that change (RQ2); whether the change goes with perceived difficulty using Arabic without AI and mediates the link from AI use (RQ3); and whether AI-use information improves out-of-sample prediction of who reports AI-attributed decreases beyond background characteristics (RQ4). Throughout, the outcome is what people *attribute* to AI, not measured language loss, and the attribution is compared with a rival one (social media, asked as a single item).

II. BACKGROUND AND HYPOTHESES

Domains and diglossia. Language use is organised by domain—family, friendship, religion, education, work—and a dominant language displaces a weaker one domain by domain, public and formal domains before private ones [9], [10]. Arabic is diglossic: Fusha (Modern Standard Arabic) carries the formal, written functions and the spoken dialect the everyday ones [11], so formal functions should be lost first. Hence **H1**: net AI-attributed change is negative for Fusha (H1a) and for work or study (H1b); and **H2**: the decrease follows the order work/study > personal matters > family (H2a) and is larger for Fusha than for the dialect domains—self-talk, personal matters, family and friends (H2b).

Why AI could matter. Adult first-language attrition is usually explained by reduced use of the first language and interference from an active second language [12], [13]. It has been studied mainly under migration; at home the first language resists: students at English-medium universities in Turkey showed no deficit on Turkish tests, although their self-rated Turkish writing fell more often than that of Turkish-medium peers [14]. Generative AI could add a new route to reduced use. Once access to a technology is equal, inequality moves from access to what people can do with it [15], and with AI people can do more in English. Hence **H3**: AI-use intensity (H3a), the share of AI use in English (H3b) and the perceived quality gap

(H3c) predict the AI-attributed decrease; **H4**: the decrease is associated with perceived difficulty using Arabic without AI; and **H5**: the decrease mediates the link from AI-use intensity to that difficulty.

Measuring attributed change. Wirtz and Pickl [16] showed that people can rate, in an online survey, how a specific cause—a major life event—changed their language use, on a scale with a true zero (“no change”). Such retrospective attributions are informative but imperfect: perceived change after life events matched measured personality change only modestly and also tracked pre-event trends [17]. This study adapts the format to AI and reads the results as attributions.

III. METHOD

A. Design, Procedure and Participants

An anonymous Google Form in Arabic (seven pages; 25 closed questions, five grids and one optional open question) was open from 30 September 2026, 17:57, to 1 October 2026, 22:15; 36 of the 40 responses arrived on the first day. Respondents were recruited through university WhatsApp groups, friends and LinkedIn. Collection was closed at 40 responses, against a planned 150 collected and about 120 analysed, before any analysis. The scheduled back-translation and pretest did not take place; only the author reviewed the wording before launch, so the Arabic wording is the instrument (Appendix) and the English used here is the author’s source wording; the advertised “about 7 minutes” was not timed. The first page described the survey as a research project at the Lebanese University, voluntary and anonymous (no names or contact details, so a response could not be withdrawn after submission) and reported only as group results; respondents confirmed being 18 or older and agreeing to take part. [ethics statement] A screening question required Arabic as first language, use of English for study or work, and weekly AI use for six months or more.

All 40 respondents consented and passed the screening question; one, who began regular AI use in 2026, may not meet the six-month criterion (the start month was not asked). Following the analysis plan, the primary sample holds migration constant: the 37 who grew up and live in the Arab region and have not moved country since 2022 (excluded: one who grew up and lives outside the Arab world, two who moved). Of the 37, 35 grew up in Lebanon and 2 in the Gulf; all live in Lebanon; 32 were under 25 and 5 were 25–34; 20 were women; 22 were studying for or held a master’s degree or higher and 14 a bachelor’s; 26 studied or worked in computing. Thirty used AI several times a day and 19 always wrote to it in English. All 40 are analysed as a sensitivity check.

B. Measures

The core grid asked: “Because of AI, do you now do each of these less or more?” (بسبب الذكاء الاصطناعي، هل أصبحت تقوم (بكلّ مما يلي أقل أم أكثر؟)، comparing now with before regular AI use, from *much less* (−2) through *no change* (0) to *much more* (+2). The eight domains are shown in Fig. 1 and, as fielded, in the Appendix. A preceding grid recorded which

language each activity is mostly done in today, with a “not applicable” option; a domain marked not applicable was set to missing. The *displacement score* is minus the mean change over the applicable domains (at least six), so higher means more AI-attributed decrease ($\alpha = 0.88$; McDonald’s $\omega = 0.88$, bootstrap 95% CI [0.75, 0.93] [18]; $n = 30$ complete cases). Its consistency may partly reflect a general tendency to blame AI, so it is read as a person-level score of attributed decrease, not as a measure of domain loss.

Perceived difficulty without AI: “Because of AI, has each of these become harder or easier for you, when you use Arabic on your own without AI?” (finding the everyday word, speaking one’s dialect fluently, saying what one means fully in Arabic; higher = harder; $\alpha = 0.78$, $\omega = 0.83$ [0.62, 0.93]; 24 of 37 scored exactly 0). *AI use*: frequency, share of work or study done with AI, breadth (number of eight uses ticked), year regular use began, the language used to write to AI (English share), and exposure to AI-generated content and the language chosen for it. *AI-use intensity* is the mean of the standardised share and breadth (as planned, frequency is descriptive only; 30 of 37 are in its top category). *Perceived quality gap*: four agreement items, one reversed ($\alpha = 0.68$, $\omega = 0.69$). Also asked: a substitution item (“Has it happened that you did something in English, although you could have done it in Arabic, because AI helps you better in English?”), the same change question for social media over the same period (one item, after the AI grid), life events during the period, expected change in each domain in two years, and English level.

Three safeguards in the analysis plan were not fielded: a reminder to choose “no change” if a change had other causes (such as university or a new job), a placebo row, and an attention check; the practice items and five planned items were also dropped before launch, so the checks built on them could not be run.

C. Analysis

The analysis follows a plan written before data collection (27 September 2026; not registered) and a study proposal (30 September). Five decisions taken after the data were seen are reported with their alternatives: returning to the plan’s core sample after a first pass on all 40 (with all 40, H4 is not supported); averaging all eight fielded domains instead of the plan’s four; adjusting for age and an English-medium start only (adding the plan’s field and AI tenure changes no conclusion); setting aside the plan’s rule for samples below 100 (H5 as the only confirmatory test, age as the only covariate, a logistic baseline as the only machine learning), except that H5 is reported as underpowered; and writing the machine-learning design after data collection.

Family A (H1–H2, within-person) used Wilcoxon signed-rank tests against zero, with exact sign tests alongside for H1a and H1b, and Page’s one-sided test for the predicted order [19], with Friedman’s test and Kendall’s W . Family B regressed the displacement score on the three AI-use measures (H3, evaluated by their joint test) and perceived difficulty on the displacement score (H4), using OLS with HC3 standard

errors and t -based inference [20], adjusted for age 25 or over and starting English-medium study or work. Holm’s correction [21] was applied within each family, and a wild bootstrap (Rademacher weights, null imposed) checked the Family B p -values [22]. H5 followed PROCESS model 4 [23] with a 5000-resample percentile bootstrap; at $n = 36$ it can detect only large paths [24]. A Firth logistic regression [25] of any net decrease checked H3 in binary form. Analyses were run in SPSS 23 [26] and Python [27], [28]; 34 statistics computed in both agree.

D. Machine Learning (RQ4)

The machine learning is exploratory: its design was written after data collection, once the class counts and one table of feature–target correlations had been seen, and a first full run was independently audited and revised before the results reported here (among other changes, a school-language feature added after the correlations were seen was removed). Targets were the number of domains with a decrease (regression), any decrease in a formal domain (work or study, or Fusha) and a decrease in each single domain with at least eight respondents in the smaller class (six domains; family and religion had too few). Features came in nested blocks: background (age 25 or over, computing field, education, English level, English-medium start), then plus AI use (seven items), then plus the quality gap; outcome items were never features. Models had fixed settings (L2 logistic or ridge regression as the primary model, random forest, gradient boosting) and were evaluated by repeated 10×5-fold cross-validation, with class imbalance handled by oversampling (SMOTENC [29], applied inside the training folds only). Paired comparisons used the Nadeau–Bengio corrected t [30]; whether a model beat chance was tested by permutation (outcome shuffled, whole pipeline refit, 500 permutations), with Holm’s correction. Three further targets named before data collection (substitution, expected decrease in two years, any perceived difficulty) were added later, with criteria for a usable model fixed before they were modelled: above chance after correction; stable over ten other random splits (mean AUC at least 0.70, none below 0.65; for a count, $Q^2 > 0$ throughout and a significant gain over the mean); not dependent on age; balanced accuracy at least 0.65. Models used scikit-learn and imbalanced-learn [31], [32].

IV. RESULTS

A. Where Respondents Report Change (RQ1)

Thirty-one of the 37 respondents reported an AI-attributed decrease in at least one domain (3.2 of their applicable domains on average). Decreases were most common in work or study (69% [53, 82], none reporting more) and writing messages and posts (57%), and least common with family and friends (19%) and for religious texts, where decreases and increases balanced (9% each); for Fusha, 16 of the 33 to whom it applied reported less and 3 more (Fig. 1). Two thirds (25 of 37, 68% [51, 80]) reported at least sometimes doing in English what they could have done in Arabic, because AI works better in English.

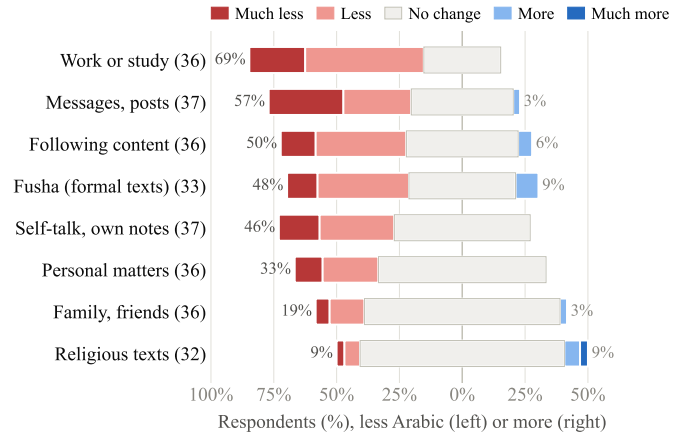


Fig. 1. AI-attributed change in Arabic use by domain (primary sample), centred on “no change”; n in parentheses excludes respondents who marked the domain not applicable.

B. Direction and Order (H1–H2)

The net change was negative for Fusha (H1a) and for work or study (H1b), and the decrease followed the predicted order work/study > personal matters > family (H2a; Table I). These results held with all 40 respondents (Holm $p \leq 0.026$). The plan’s own test of H1b and H2a, restricted to people whose work or study setting did not change, could not be run as intended: only eight respondents met a proxy for it (work or study: 3 less, 0 more, $p = 0.25$; order: $n = 7$, Page one-sided $p = 0.062$). The decrease for Fusha was not larger than for the dialect domains (H2b).

C. Predictors and Correlates (H3–H5)

The three AI-use measures did not jointly predict the displacement score, and a Firth logistic regression of any net decrease agreed (all three $p \geq 0.26$). AI-use intensity and the perceived quality gap were unrelated to it. The English share of AI use was positive in every specification but survived Holm’s correction only when the score was restricted to the plan’s four domains or to the dialect domains (Table II); it is confounded with English proficiency ($\rho = 0.56$; $p = 0.22$ once proficiency is controlled) and is not significant once social-media attribution is controlled ($p = 0.098$). The largest coefficient belonged to age: the five respondents aged 25 or over reported almost no AI-attributed change (four none at all; $b = -0.65$, $p = 0.001$).

Respondents who attributed more decrease to AI also reported more AI-attributed difficulty using Arabic without AI (H4). The association was positive in every specification of the primary sample but significant after correction in only about half of those run (five of ten; two of the seven in Table II); removing any one of three respondents pushes it above the Holm threshold, and with all 40 it disappears ($b = 0.12$), driven by one respondent outside the primary sample. Both measures are attributions on the same page, so H4 is an association between two AI attributions, not evidence that reduced use erodes ability. AI-use intensity did not predict

TABLE I
CONFIRMATORY TESTS (PRIMARY SAMPLE)

Hypothesis	<i>n</i>	Result	Test	Holm <i>p</i>	Verdict
<i>Family A: direction and order of AI-attributed change</i>					
H1a Fusha	33	16 less, 3 more	$z = -3.04, r = -0.53$	0.005	supported
H1b Work or study	36	25 less, 0 more	$z = -4.56, r = -0.76$	<0.001	supported
H2a Work/study > personal > family	34	means 0.97, 0.41, 0.15	Page; $\chi^2(2) = 30.16, W = 0.44$	<0.001	supported
H2b Fusha > dialect domains	33	difference +0.12	$z = 1.14$	0.26	not supported
<i>Family B: H3 predicts the displacement score; H4 predicts difficulty from it (<i>n</i> = 36, HC3)</i>					
H3 (joint)	36	three predictors	$F(3, 30) = 1.46, p = 0.24$		not supported
H3a AI-use intensity		$b = -0.01 [-0.21, 0.19]$	$p = 0.95$	0.95	
H3b English share of AI use		$b = 0.22 [0.005, 0.43]$	$p = 0.045$	0.14	
H3c Perceived quality gap		$b = -0.12 [-0.40, 0.17]$	$p = 0.41$	0.82	
H4 Difficulty without AI	36	$b = 0.44 [0.14, 0.75]$	$p = 0.006$	0.024	supported ^a
H5 Mediation	36	indirect $-0.02 [-0.14, 0.04]$	percentile bootstrap		not supported ^b

Wilcoxon tests against zero (for H1a and H1b, exact sign tests agree); H2a: Page’s one-sided test, complete-case means. Family B adjusts for age 25 or over and an English-medium start; brackets are 95% CIs. ^aHolm-significant in about half of the specifications, not after removing any one of three influential respondents, and not with all 40 (Table II); wild bootstrap Holm $p = 0.0496$. ^bUnderpowered.

TABLE II
ENGLISH SHARE (H3B) AND H4 UNDER ALTERNATIVE SPECIFICATIONS

Specification	English share		H4	
	<i>b</i>	Holm <i>p</i>	<i>b</i>	Holm <i>p</i>
Primary specification	0.22	0.136	0.44	0.024
Plan’s 4-domain score	0.30	0.027	0.36	0.067
Dialect domains only	0.29	0.036	0.30	0.141
Age only (plan’s small-sample rule)	0.22	0.114	0.40	0.087
No covariates	0.19	0.273	0.35	0.176
Adding English proficiency	0.15	0.657	0.41	0.018
Without straight-line responders	0.16	0.285	0.41	0.285
All 40 respondents	0.30	0.189	0.12	1.000

Holm’s correction across the four Family B tests within each specification. Straight-line responders answered “much less” on every change row; only one of the two has a displacement score.

the displacement score ($a = -0.06, p = 0.55$) and the indirect effect was close to zero, so H5 was not supported. Exploratory, as planned: switching between Arabic and English showed no indirect effect, and among the 28 who wrote Fusha before AI, finding formal writing harder went with more attributed decrease ($\rho = 0.57, p = 0.002$; both AI attributions).

D. AI or Technology in General?

AI-attributed and social-media-attributed decreases went together ($\rho = 0.56$). Respondents’ single social-media rating showed more decrease than their AI ratings averaged over the domains (+0.72 vs. +0.49, both scored like the displacement score; $p = 0.014$) and as much as their AI ratings for the two formal domains (+0.72 each); since one overall item is compared with domain averages, this brackets the comparison rather than settling it. In an exploratory split, among the 15 who did not blame social media the AI-attributed decrease was small and not significant (+0.14, $p = 0.25$); among the 21 who did, it was +0.74. Life transitions also coincided with the AI period: 27 of the 32 respondents under 25 started university, graduated, began a new job or started studying or working in English, and those who started university reported

larger work/study decreases (mean change -1.31 vs. -0.70 on the -2 to $+2$ scale, Mann–Whitney $p = 0.016$; exploratory).

E. Prediction (RQ4)

Adding AI-use information never reliably improved prediction (Table III): for the number of domains with a decrease, out-of-sample Q^2 changed by $-0.27 [-0.62, 0.08]$, and for any formal-domain decrease AUC changed by $-0.01 [-0.34, 0.32]$ (Holm $p = 0.26$ and 0.94); the change was negative on 10 of 10 and 9 of 10 further random splits, and AI use alone predicted neither target better than chance. The corrected test could detect only gains of about 0.5, but a compact block of the two H3 measures bounds the gain for the count ($\Delta Q^2 = -0.03 [-0.14, 0.09]$). Background characteristics carried some signal about the number of domains with a decrease ($Q^2 = 0.14$, permutation $p = 0.004$), but their advantage over predicting the mean was not significant ($+0.20 [-0.15, 0.56]$, $p = 0.26$), and the signal vanished without the age variable ($Q^2 = -0.17, p = 0.31$); the same held for writing (AUC 0.76, falling to 0.51 without age). Random forests and gradient boosting never beat the linear model by the decision rule’s margin (0.08 AUC or 0.05 Q^2 , corrected $p < 0.05$). No further target came close to the usability criteria; across all 11 targets, the smallest Holm-adjusted p for background plus AI was 0.42.

Oversampling did not change this (Table IV; a post hoc check). Six strategies applied inside the training folds gave no reliable gain over no oversampling (best +0.04 AUC; Holm $p = 1.00$ across 24 comparisons), and the best valid result fell to 0.57 without the age variable (permutation $p = 0.30$). Applied the invalid way, to all respondents before cross-validation, every target looked predictable, including substitution, for which no signal was detected (AUC 0.42 without oversampling).

V. DISCUSSION

Most respondents (31 of 37) reported that AI has reduced their use of Arabic in at least one domain, and they place

TABLE III
CROSS-VALIDATED PREDICTION (LINEAR MODELS, PRIMARY SAMPLE)

Target (<i>n</i> yes/no; metric)	Background	+ AI use	AI use only	Δ AI use [95% CI]	Usable
Domains with a decrease (count; Q^2)	0.14 ^a	-0.13	-0.36	-0.27 [-0.62, 0.08]	no
Any formal-domain decrease (27/9; AUC)	0.71	0.70	0.52	-0.01 [-0.34, 0.32]	no
Writing messages and posts (21/16; AUC)	0.76 ^a	0.58	0.39	-0.18 [-0.35, -0.01]	no
<i>Further targets, criteria fixed before they were modelled</i>					
Substitution (25/12; AUC)	0.29	0.32	0.39	+0.03 [-0.26, 0.33]	no
Expected decrease in two years (count; Q^2)	0.02	-0.61	-0.49	-0.62 [-1.18, -0.06]	no
Any difficulty without AI (12/25; AUC)	0.57	0.54	0.52	-0.03 [-0.27, 0.21]	no

Means over repeated 10×5 -fold cross-validation; chance is AUC 0.50 and $Q^2 = 0$ (negative: worse than predicting the mean). Δ : background plus AI use minus background, Nadeau-Bengio CI. Usable: all four criteria of Section III-D (for the first three targets, applied after their results were known). ^aAbove chance after Holm’s correction only through the age variable (five respondents aged 25 or over). Other single domains: background AUC 0.40–0.68, AI-use changes -0.15 to +0.13, none significant. AUCs below 0.50 reflect no signal in a small sample, and negative Δ intervals overfitting seven features to 37 people; neither is a finding.

TABLE IV
OVERSAMPLING INSIDE VERSUS OUTSIDE THE CROSS-VALIDATION (AUC;
BACKGROUND PLUS AI USE)

Target	Valid: training folds only			Invalid
	None	SMOTENC	Best of six	Before split
Any formal-domain decrease	0.69	0.70	0.73	0.96
Writing messages and posts	0.60	0.58	0.64	0.86
Substitution	0.42	0.32	0.43	0.84
Any difficulty without AI	0.51	0.54	0.54	0.89

Strategies: class weights, random duplication, SMOTENC, SMOTENC amplified (both classes grown to five times the majority), SMOTENC with Tomek links, five jittered copies of everyone. Invalid: the highest of the four that add cases, applied before cross-validation; it measures recognition of synthetic copies, not prediction.

the reduction where theory expects it: in work and study, writing, following content and Fusha, and hardly at all with family or in religion. The formal part of this profile matches reports under English-medium education [4], [5]; here it is attributed to a new cause and extends to messages and media content. The order work/study > personal > family is also the order of where AI is used, so it does not by itself confirm domain theory, and H2b, which would separate the two, was not supported. The attributions seem to follow tasks rather than varieties—less Arabic mainly in the activities people now do with AI—although a null H2b at $n = 33$ cannot rule out a difference between varieties.

The study cannot show that AI causes this. The outcome is the respondents’ own attribution, collected on a page titled “Has AI changed your Arabic?” without a placebo row or a reminder to exclude other causes. The attribution is shared with social media, which respondents blamed about as much on a single overall item. For most young respondents the AI period also coincides with the move from school to university or work, and such life events are known to shape perceived language change [16]; because the planned test restricted to people whose setting did not change could not be run, a change of setting remains an open rival explanation, especially for the work/study decrease (H1b). And the evidence a non-experimental design would normally offer for a cause—more

decrease among heavier or more English-routed AI users—is weak: intensity was unrelated to the decrease, the English share survived correction only under two alternative scorings and not once English proficiency or social-media attribution was controlled, and AI-use information added no detectable predictive gain. A plausible reading is that AI joins a broader sense that technology and English are eroding Arabic, rather than acting as a separate force.

The machine-learning results are a substantive answer, not a failure. With 37 people, background and self-reported AI use gave no usable prediction of who reports AI-attributed decreases, and AI-use information added no detectable gain, though the test could detect only gains of about 0.5; the only predictable contrast is the five respondents aged 25 or over, who report almost no change. Oversampling cannot add information: done correctly it made no difference, and done before the split it manufactured AUCs above 0.80 even for a target with no detectable signal—an error that has produced near-perfect but spurious results elsewhere [33], [34].

Limitations. The sample is small (37 analysed against a planned 120), recruited through the author’s university and personal networks in a single burst, and young, computing-trained and Lebanese; percentages describe this sample, not bilinguals in general. The instrument was fielded without back-translation or pretest, so how respondents understood “Arabic” (for example, whether Arabizi counts) and “because of AI” is unknown, and change was rated retrospectively, over roughly two to four years for most respondents (35 of 37 began regular AI use in 2024 or earlier). The analysis plan was not registered, and several decisions, including the whole machine-learning design, were taken after the data were seen (Section III-C). At this sample size the smallest correlation detectable with 80% power is about 0.45, so every null result here is weak evidence. *Future work:* a larger sample with the planned safeguards and counterbalanced AI and social-media questions would test whether the AI attribution is specific; longitudinal designs and behavioural records of the language people use with AI would move from attribution to measured change.

VI. CONCLUSION

Is generative AI displacing Arabic? These bilinguals report that it has reduced their Arabic in work or study, in writing and in Fusha, but in this small sample no measure of how much or how they use AI reliably predicted who reports it, and the attribution cannot be separated from a broader sense that technology is eroding Arabic. Answering the question needs larger samples and measures of behaviour, not only of belief.

DATA AND CODE AVAILABILITY

The instrument (Arabic and English), analysis code and outputs are at <https://github.com/hadigghazi/genai-arabic-displacement>. Individual responses are not shared, as participants were told results would be reported as group results.

APPENDIX: KEY ITEMS AS FIELDDED

Because of AI, do you now do each of these less or more?	بسبب الذكاء الاصطناعي، هل أصبحت تقوم بكل مما يلي أقل أم أكثر؟
Using Arabic in my work or studies	استخدام العربية في عملي أو دراسي
Writing messages or posts in Arabic	كتابة الرسائل أو المنشورات بالعربية
Using Arabic when I talk to myself or write my own notes	استخدام العربية عندما أحدث نفسي أو أكتب ملاحظاتي الخاصة
Using Arabic for personal matters	استخدام العربية في الحديث عن أموري الشخصية
Using Arabic with my family or friends	استخدام العربية مع عائلتي أو أصدقائي
Reading or writing Fusha (news, documents, articles)	قراءة القصص أو كتابتها (أخبار، وثائق، مقالات)
Reading or listening to religious texts	قراءة نصوص دينية أو الاستماع إليها
Following content in Arabic (videos, articles)	متابعة محتوى بالعربية (فيديوهات، مقالات)

REFERENCES

- [1] K. Kubrak, A. El-Moselhy, A. Alsulami, R. Altuwaim, H. I. Fawaz, and F. Alsaby, "Arabic prompts with English tools: A benchmark," in *Proc. 2025 IEEE Int. Conf. Big Data (BigData)*, 2025, pp. 5452–5461.
- [2] M. Sallam, K. Al-Mahzoum, O. Alshuaib *et al.*, "Language discrepancies in the performance of generative artificial intelligence models: An examination of infectious disease queries in English and Arabic," *BMC Infectious Diseases*, vol. 24, no. 1, 2024, art. no. 799.
- [3] A. Al-Bataineh, "Language policy in higher education in the United Arab Emirates: Proficiency, choices and the future of Arabic," *Language Policy*, vol. 20, no. 2, pp. 215–236, 2021.
- [4] T. I. Masri, "A critical examination of the Arab undergraduate students' perceptions of their academic Arabic proficiency in three EMI universities in the UAE," *English Language Teaching*, vol. 12, no. 4, p. 21, 2019.
- [5] E. Mustafawi, K. Shaaban, T. Khwaileh, and K. Ata, "Perceptions and attitudes of Qatar University students regarding the utility of Arabic and English in communication and education in Qatar," *Language Policy*, vol. 21, no. 1, pp. 75–119, 2022.
- [6] M. Mehio, "Upholding and resisting English language ideologies: Identity negotiation of five Sunni Muslims in Lebanon," *Journal of Language, Identity & Education*, vol. 25, no. 5, pp. 2132–2146, 2026.
- [7] M. Saarela, S. Gunasekara, and P. Kumarage, "A meta-analysis of generative AI effects on language proficiency and affective–cognitive outcomes in language learning," *Discover Computing*, vol. 29, no. 1, 2026, art. no. 116.
- [8] A. Bouzayenne, "Language choice and AI-mediated code-switching in Tunisians' interactions with ChatGPT: Age-related and task-domain variation," *Dialogica*, vol. 2, no. S1, pp. 33–54, 2026.
- [9] J. A. Fishman, "Who speaks what language to whom and when?" *La Linguistique*, vol. 1, no. 2, pp. 67–88, 1965.
- [10] —, "Domains and the relationship between micro- and macrosociolinguistics," in *Directions in Sociolinguistics: The Ethnography of Communication*, J. J. Gumperz and D. Hymes, Eds. New York, NY, USA: Holt, Rinehart and Winston, 1972, pp. 435–453.
- [11] C. A. Ferguson, "Diglossia," *Word*, vol. 15, no. 2, pp. 325–340, 1959.
- [12] A. Chaouch-Orozco and F. Martín-Villena, "Network science reveals the early signs of L1 lexical attrition: Introducing the Lexical Attrition Foundation (LeAF) framework," *Bilingualism: Language and Cognition*, vol. 28, no. 1, pp. 43–53, 2025.
- [13] F. Gallo, B. Bermúdez-Margaretto, A. Malyshevskaya *et al.*, "Lexico-semantic attrition of native language: Evidence from Russian–Hebrew bilinguals," *Language Learning*, vol. 75, no. 3, pp. 771–800, 2025.
- [14] A. Ayçiçeği-Dinn, S. Şişman-Bal, and C. L. Caldwell-Harris, "Does attending an English-language university diminish abilities in the native language? Data from Turkey," *Applied Linguistics*, 2015, advance online publication.
- [15] E. Hargittai, "Second-level digital divide: Differences in people's online skills," *First Monday*, vol. 7, no. 4, 2002.
- [16] M. A. Wirtz and S. Pickl, "Major life events as drivers of perceived linguistic change across adulthood," *Language Variation and Change*, vol. 37, no. 1, pp. 87–110, 2025.
- [17] T. Schwaba, J. J. A. Denissen, M. Luhmann, C. J. Hopwood, and W. Bleidorn, "Subjective experiences of life events match individual differences in personality development," *Journal of Personality and Social Psychology*, vol. 125, no. 5, pp. 1136–1156, 2023.
- [18] T. J. Dunn, T. Baguley, and V. Brunnsden, "From alpha to omega: A practical solution to the pervasive problem of internal consistency estimation," *British Journal of Psychology*, vol. 105, no. 3, pp. 399–412, 2014.
- [19] E. B. Page, "Ordered hypotheses for multiple treatments: A significance test for linear ranks," *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 216–230, 1963.
- [20] J. S. Long and L. H. Ervin, "Using heteroscedasticity consistent standard errors in the linear regression model," *The American Statistician*, vol. 54, no. 3, pp. 217–224, 2000.
- [21] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [22] R. Davidson and E. Flachaire, "The wild bootstrap, tamed at last," *Journal of Econometrics*, vol. 146, no. 1, pp. 162–169, 2008.
- [23] A. F. Hayes, *Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach*, 3rd ed. New York, NY, USA: Guilford Press, 2022.
- [24] M. S. Fritz and D. P. MacKinnon, "Required sample size to detect the mediated effect," *Psychological Science*, vol. 18, no. 3, pp. 233–239, 2007.
- [25] D. Firth, "Bias reduction of maximum likelihood estimates," *Biometrika*, vol. 80, no. 1, pp. 27–38, 1993.
- [26] *IBM SPSS Statistics for Windows, Version 23.0*, IBM Corp., Armonk, NY, USA, 2015.
- [27] S. Seabold and J. Perktold, "Statsmodels: Econometric and statistical modeling with Python," in *Proc. 9th Python in Science Conf.*, 2010, pp. 92–96.
- [28] P. Virtanen, R. Gommers, T. E. Oliphant *et al.*, "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, no. 3, pp. 261–272, 2020.
- [29] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [30] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003.
- [31] F. Pedregosa, G. Varoquaux, A. Gramfort *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [32] G. Lemaitre, F. Nogueira, and C. K. Aridas, "Imbalanced-learn: A Python toolbox to tackle the curse of imbalanced datasets in machine learning," *Journal of Machine Learning Research*, vol. 18, no. 17, pp. 1–5, 2017.
- [33] G. Vandewiele, I. Dehaene, G. Kovács *et al.*, "Overly optimistic prediction results on imbalanced data: A case study of flaws and benefits when applying over-sampling," *Artificial Intelligence in Medicine*, vol. 111, 2021, art. no. 101987.
- [34] M. S. Santos, J. P. Soares, P. H. Abreu, H. Araujo, and J. Santos, "Cross-validation for imbalanced datasets: Avoiding overoptimistic and overfitting approaches," *IEEE Computational Intelligence Magazine*, vol. 13, no. 4, pp. 59–76, 2018.